

Haseeb Chatter

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BSCS-4E

Ru

Q:- $\Sigma = \{0, 1\}$

① $\{w: w \text{ contain at least three 1s}\}$

$$(0+1)^* 1 (0+1)^* 1 (0+1)^* 1 (0+1)^*$$

② $\{w: w \text{ contain at least two 1s and at most one 0}\}$

$$1^* 0 1^* 1^* + 1^* 1^* 0 1^* + 1^* 1^*$$

③ $\{\text{Contain even no. of 0's and exactly two 1s}\}$

$$(0^* 1 0^* 1 (0^* 0)^* + 0^* 0 1^* 0)^*$$

(4) $\{w: w \text{ contain exactly two } 0\text{'s}$
and at least two $1\text{'s}\}$

$1^* 00111^*$

(5) $\{w: w \text{ contain even no. of } 0\text{'s}$
and each 0 is followed by at least
 $1\}$

$(1^* 01 + 1^*)^*$

(6) $\{w: w \text{ every odd position } w \text{ is } 1\}$

$1(010)^*$

(7) $\{w: w \text{ any string except '00' and '11'}\}$

$(01)^* 0110(01)^*$

Q: when $\Sigma = \{x, y\}$

⑧ $L = \{w \mid \text{third position is 'y'}\}$

$$x^n (xy)(ny) y (ny)^n$$

⑨ $L = \{w \mid \text{length of } w \text{ is multiple of three}\}$

$$(xy)(xy)(ny)^n$$

⑩ All words whose second last symbol is 'x'.

$$(ny)^n y^* y^* x(ny)$$

$$(ny)^n y^* x(ny)$$